# **Recitation (3)** (Electrostatics)

Assume in all problems a free space medium $\varepsilon = \varepsilon_0$

**Problem 3.1:**

1. Point charges $Q_1$ and $Q_2$ are, respectively, located at (4,0, -3) and (2,0,1). If $Q_2 = 4nC$, find $Q_1$if:
   a) $\vec{\boldsymbol{E}}$ at (5, 0, 6) has no $z$-component.
   b) The force on a test charge at $(5,0,6)$ has no $x$-component.
2. A $100\text{nC}$ point charge is located at $A(-1,1,3)$ in free space, Find the locus of all points $P(x,y,z)$ at which $E_x = 500\ \text{V/m}$.

**Problem 3.2:**

1. Find the electric field on the $z$ axis at $(0,0,h)$ produced by an annular ring of uniform surface charge density $\rho_s$ in free space. The ring occupies the region $z = 0, a \le \rho \le b, 0 \le \phi \le 2\pi$ in cylindrical coordinates.
2. Obtain the field of an infinite uniform sheet charge by taking appropriate limits.

**Problem 3.3:**

A ring placed along $y^2 + z^2 = 4, x = 0$ carries a uniform charge of $5\mu\text{C/m}$.

(a) Find D at $P(3,0,0)$.

(b) If two identical point charges $Q$ are placed at $(0,-3,0)$ and $(0,3,0)$ in addition to the ring, find the value of $Q$ such that $\text{D} = 0$ at $P$.

**Problem 3.4:**

1. A 10nC charge is uniformly distributed over a spherical shell $r = 3$ cm, and a -5nC charge is uniformly distributed over another spherical shell $r = 5$ cm. Find $\vec{\boldsymbol{D}}$ for regions $r < 3$ cm, $3\text{ cm} < r < 5$ cm, $r > 5$ cm.
2. In free space, $V = \frac{1}{r^3}\sin\theta\cos\phi$. Find $\vec{\boldsymbol{E}}$ and $\vec{\boldsymbol{D}}$ at $(1, 30°, 60°)$, Show that $\vec{\boldsymbol{E}}$ is conservative.